Precision [%]

30 40 50 60 70 80 90 100

Scenario	Application	JSON & application/dns-message	CBOR & application/dns-message	CBOR & application/dns+cbor	JSON & application/dns+cbor	JSON & application/dns-message	CBOR & application/dns-message	CBOR & application/dns+cbor	JSON & application/dns+cbor
Split server, direct	CoAP	97.6	96.3	81.3	68.2	93.1	100.0	94.7	90.3
	OSCORE	79.4	91.4	67.2	66.7	73.6	82.9	74.8	82.2
	Onion OSCORE	74.6	74.9	68.3	68.5	82.4	86.1	87.3	85.7
	CoAPS	70.5	74.8	65.0	69.1	73.4	91.8	73.7	67.9
	HTTPS	100.0	100.0	100.0	100.0				
Split server, via proxy	CoAP	95.4	95.1	70.7	70.2	93.6	94.3	67.2	70.3
	OSCORE	78.0	79.2	69.1	70.7	76.5	69.3	72.2	48.6
	Onion OSCORE	67.3	68.4	63.7	65.5	45.1	49.5	34.8	32.4
	CoAPS	67.7	68.7	63.7	66.5	51.2	55.0	41.7	37.5
	HTTPS	66.4	60.6	73.5	55.2				
Single server, via proxy	CoAP	98.4	99.2	70.7	70.7	89.7	89.6	53.1	47.1
	OSCORE	68.2	69.3	63.8	66.7	50.1	53.6	43.0	39.7
	Onion OSCORE	67.5	67.7	64.3	65.6	43.9	48.7	35.0	31.6
	CoAPS	67.8	68.3	64.1	66.8	49.8	54.1	40.9	38.4
	HTTPS	65.5	72.5	56.4	54.7				
Single server, direct	CoAP	85.3	84.2	70.8	72.0	77.9	79.9	47.0	41.7
	OSCORE	69.0	69.2	63.3	66.7	46.3	49.8	36.5	33.6
	Onion OSCORE	68.6	69.2	63.7	66.2	46.2	52.8	36.8	33.1
	CoAPS	70.5	70.2	63.8	66.9	51.7	55.4	37.7	33.6
	HTTPS	59.6	60.5	54.0	55.4				

Scenario	Application	JSON & application/dns-message	CBOR & application/dns-message	CBOR & application/dns+cbor	JSON & application/dns+cbor	JSON & application/dns-message	CBOR & application/dns-message	CBOR & application/dns+cbor	JSON & application/dns+cbor
Split server, direct	CoAP								
	OSCORE	65.9	67.5	67.2	65.3	100.0	100.0	87.4	87.1
	Onion OSCORE	65.6	66.1	65.8	65.1	100.0	100.0	88.4	85.9
	CoAPS	62.2	61.5	68.3	66.9	85.8	86.7	81.3	90.9
	HTTPS								
Split server, via proxy	CoAP								
	OSCORE	64.6	66.1	65.7	64.3	80.2	66.9	71.8	69.8
	Onion OSCORE	62.1	62.6	62.4	62.3	36.0	39.9	32.8	29.7
	CoAPS	61.2	60.8	65.8	64.7	55.6	59.2	44.5	41.2
	HTTPS								
Single server, via proxy	CoAP								
	OSCORE	61.1	63.1	63.5	62.8	68.2	61.1	49.1	44.8
	Onion OSCORE	62.4	62.2	62.0	61.7	35.6	39.2	32.5	29.1
	CoAPS	60.9	60.9	65.1	65.1	54.9	58.2	44.7	40.1
	HTTPS								
Single server, direct	CoAP								
	OSCORE	61.3	63.0	63.6	62.3	53.7	53.8	42.9	40.2
	Onion OSCORE	61.3	62.6	63.2	62.0	60.0	54.7	42.2	40.5
	CoAPS	60.6	60.5	65.2	64.3	62.0	65.9	43.2	39.6
	HTTPS								

Scenario	Application	"Ethernet" $B = 1024$	SCHC (split) $B = 1024$	SCHC (min) $B = 1024$	SCHC (peer-based) $B = 1024$	"Ethernet" $B = 64$	SCHC (split) $B = 64$	SCHC (min) $B = 64$	SCHC (peer-based) $B = 64$
Split server, direct	CoAP								
	OSCORE	79.4	65.9	65.9	61.5	73.6	100.0	100.0	52.7
	Onion OSCORE	74.6	65.6	65.3	61.7	82.4	100.0	100.0	61.2
	CoAPS	70.5	62.2	63.9	60.0	73.4	85.8	88.7	74.3
	HTTPS								
Split server, via proxy	CoAP								
	OSCORE	78.0	64.6	61.2	64.6	76.5	80.2	76.7	80.2
	Onion OSCORE	67.3	62.1	62.1	62.1	45.1	36.0	36.0	36.0
	CoAPS	67.7	61.2	61.2	61.2	51.2	55.6	55.6	55.6
	HTTPS								
Single server, via proxy	CoAP								
	OSCORE	68.2	61.1	61.1	61.1	50.1	68.2	68.2	68.2
	Onion OSCORE	67.5	62.4	62.4	62.4	43.9	35.6	35.6	35.6
	CoAPS	67.8	60.9	60.9	60.9	49.8	54.9	54.9	54.9
	HTTPS								
Single server, direct	CoAP								
	OSCORE	69.0	61.3	61.3	61.3	46.3	53.7	53.7	53.7
	Onion OSCORE	68.6	61.3	61.3	61.3	46.2	60.0	60.0	60.0
	CoAPS	70.5	60.6	60.6	60.6	51.7	62.0	62.0	62.0
	HTTPS								

Link Layer Mode & Block Size B [bytes]